1 Correct Answer: C. Computer Organization

Computer Organization refers to how the features are implemented; it describes the physical realization and interconnectivity of the computer components (like control signals, memory types, and peripherals). Architecture refers to the attributes visible to the programmer (like instruction set and number of registers).

2 Correct Answer: C. Instructions (Program) and Data.

The Von Neumann Architecture is also known as the Stored-Program concept. Its defining feature is the use of a single memory to hold both the program instructions and the data that the program operates on. This leads to the "Von Neumann bottleneck."Shutterstock

3 Correct Answer: D. Parallelism

Parallelism involves the simultaneous execution of different parts of a program on multiple processors to reduce total execution time. Concurrency refers to the ability of a system to handle multiple tasks seemingly at once, often by time-slicing on a single core.

4 Correct Answer: D. SISD

SISD stands for Single Instruction, Single Data. This classification describes a traditional uniprocessor computer where a single control unit fetches and executes instructions sequentially on a single stream of data.

5 Correct Answer: C. UMA

UMA stands for Uniform Memory Access. In a UMA architecture, all processors can access any word of main memory in the same amount of time. This is characteristic of Symmetric Multiprocessor (SMP) systems.

6 Correct Answer: D. Instruction-Level Parallelism (ILP)

Instruction-Level Parallelism (ILP) exploits the ability to execute multiple machine instructions concurrently within a single clock cycle, primarily through hardware mechanisms like pipelining and multiple execution units (superscalar design).

7 Correct Answer: C. Arithmetic Logic Unit (ALU)

The Arithmetic Logic Unit (ALU) is the digital circuit within the CPU that performs the fundamental arithmetic and logic operations.

8 Correct Answer: C. Load Balancing

Load Balancing is the process of distributing computation and communication workload across the available processors and cores to maximize throughput and minimize response time, ensuring efficient use of all resources.

9 Correct Answer: B. General-Purpose Graphics Processing Unit

GPGPU stands for General-Purpose Graphics Processing Unit. This term refers to the use of a GPU, which was originally designed for graphics, to perform highly parallel computations traditionally handled by the CPU.

10 Correct Answer: C. Resource Manager / Job Scheduler

The Resource Manager (or Job Scheduler) in an HPC cluster (such as SLURM or PBS) is a specialized software system whose function is to allocate the cluster's compute nodes and resources to user jobs based on defined policies and availability.

11 Correct Answer: B. Execute multiple independent instructions in parallel in the same clock cycle using multiple execution units.

A Superscalar processor design incorporates multiple execution units (like ALUs, Load/Store units) and a mechanism to dispatch and execute several instructions simultaneously within the same clock cycle, exploiting Instruction-Level Parallelism (ILP).

12 Correct Answer: C. The number of available general-purpose registers and the instruction formats.

The ISA defines the contract between the hardware and the software. It specifies the visible components to the assembly language programmer, including the set of instructions, instruction formats, the number of registers, and memory addressing modes.

13 Correct Answer: B. Provide the mechanism for the CPU, memory, and I/O modules to communicate.

The Interconnection Structure (often a bus or system of buses) is the set of physical pathways and protocols that enables the exchange of data and control information between the computer's major components: the processor, main memory, and I/O modules.

14 Correct Answer: C. SIMD

SIMD stands for Single Instruction, Multiple Data. Vector processors and modern SIMD extensions allow one instruction to be broadcast and executed on an array or vector of data elements simultaneously, which perfectly matches the SIMD model.

15 Correct Answer: C. Access time to memory is slower if the data is located on a remote node's local memory.

In a NUMA system, memory is physically partitioned and local memory access is faster for the physically attached processor. The trade-off is that remote memory access (accessing memory attached to a different processor node) takes a significantly longer time than local access, resulting in non-uniform latency.

16 Correct Answer: B. The output of one task is used as the input for the other task.

A Data Dependency (specifically a Read-After-Write or true dependency) means that one task (the consumer) must wait for the result produced by a preceding task (the producer). Such dependencies severely restrict the ability to execute the two tasks in parallel.

17 Correct Answer: C. A single control unit directing a set of identical processing elements, all executing the same instruction simultaneously on different data.

An Array Processor is a classic SIMD machine. It features a central control unit that broadcasts a single instruction to many processing elements (PEs). Each PE executes that instruction concurrently but operates on a different piece of data.

18 Correct Answer: B. Process ID (PID) and Address Space (Code, Data, and Heap).

Threads are often called "lightweight processes" because they share the memory space, code section, data section, and resources (like file handles) of their parent process. They have their own stack, program counter, and register state.

19 Correct Answer: B. Data Decomposition

Data Decomposition (or Domain Decomposition) is the strategy of partitioning the large dataset associated with a problem (e.g., a large matrix or grid) into smaller, manageable sub-problems, each assigned to a different core or node for parallel processing.

20 Correct Answer: C. To allocate and manage compute resources for user jobs.

SLURM (Simple Linux Utility for Resource Management) is the most widely used open-source workload manager/job scheduler in HPC clusters. Its core function is

to allocate exclusive or shared access to compute nodes and manage the queue of jobs waiting to run.

21 Correct Answer: C. SIMD / MIMD Hybrid

Modern GPUs operate as a hybrid. At the highest level (thread blocks and warps/wavefronts), they operate like MIMD (multiple independent instruction streams). However, within a streaming multiprocessor (SM), the threads in a single warp/wavefront execute the same instruction simultaneously on different data, making the fundamental execution unit behave as SIMD.

22 Correct Answer: C. T = IC X CPI X T_c

The fundamental CPU Performance Equation is:

$$T_{\text{CPU}} = \text{IC} \times \text{CPI} \times T_c$$

Where T_{CPU} is the total execution time, IC is the instruction count, CPI is the average cycles per instruction, and T_c is the clock cycle time ($1 / \text{Frequency}$).

23 Correct Answer: D. It has built-in mechanisms to handle data streaming and memory access for large, contiguous data sets efficiently.

Dedicated vector processors have specialized vector registers and, crucially, vector-optimized memory access hardware (strided access, banking) that efficiently loads and stores large, contiguous data blocks, minimizing the overhead typically associated with cache misses and non-contiguous data access in scalar processors.

24 Correct Answer: B. Hybrid Memory Architecture

A Hybrid Memory Architecture combines features of both distributed and shared memory. The most common HPC hybrid uses distributed memory across nodes (using MPI) and shared memory within each node (using threads like OpenMP). The description provided, while slightly different, still fits the hybrid definition of combining distinct memory models.

25 Correct Answer: B. Concurrency, primarily achieved through Task-Level Parallelism

Running multiple unrelated applications is primarily a demonstration of Concurrency, where the OS handles many independent tasks. Since the tasks are distinct applications, the parallelism is achieved by assigning these whole tasks to different cores, which is Task-Level Parallelism (TLP).

26 Correct Answer: B. To ensure that all threads reach a specific point in the code before any of them proceed further.

A Synchronization Barrier is a mechanism used to coordinate the execution of multiple threads in parallel regions. It ensures that no thread continues execution past the barrier point until all threads in the team have also reached that point.

27 Correct Answer: C. Instruction-Level Parallelism

Pipelining allows the execution of multiple instructions to be overlapped in different stages (Fetch, Decode, etc.). This overlapping of instruction execution in time is a fundamental hardware technique for exploiting Instruction-Level Parallelism (ILP) within a single core.

28 Correct Answer: D. MIMD

MIMD stands for Multiple Instruction, Multiple Data. It is the most general parallel architecture, where multiple independent processors can simultaneously execute different instruction sequences on different data streams. Modern supercomputers and multi-core systems are typically classified as MIMD.

29 Correct Answer: B. Ability to reconfigure the hardware logic itself to exactly match the computational task.

An **FPGA** contains a configurable array of logic blocks. Its main advantage is that it can be **reprogrammed** (reconfigured) to implement custom, highly parallel, and specialized hardware logic circuits specifically tailored for a target application, allowing for extreme efficiency (often surpassing CPUs and GPUs for specific tasks).

30 Correct Answer: B. All processors must be identical and have equal access to all peripheral devices and memory.

In an SMP system, multiple identical processors are connected to a single shared main memory (UMA model). Crucially, the system is symmetric: all processors share the same operating system, and have uniform access to all I/O devices and memory addresses.

31 Correct Answer: D. Modified (M)

The Modified (M) state in the MESI protocol signifies that the cache line is valid, the block is only present in the local cache (i.e., exclusive), and its value is different

from (more recent than) the value in main memory. This "dirty" status means the cache must write the data back before another processor can use it.

32 Correct Answer: A. 10 times

Amdahl's Law states that the maximum speedup (S) is limited by the serial fraction (F_{serial}):

$$S_{\text{max}} = \frac{1}{F_{\text{serial}}}$$

- Total Time = 100s.
- Serial Time = 10s.
- Serial Fraction (F_{serial}) = $10 / 100 = 0.1$.
- Maximum Speedup = $1 / 0.1 = \mathbf{10}$.

The parallel portion (90s) is fully sped up (approaches 0s), but the 10s serial portion remains, limiting the total speedup to 10.

33 Correct Answer: D. Warp (or Wavefront in AMD)

A Warp (NVIDIA) or Wavefront (AMD) is the fundamental unit of execution scheduling on a GPU's Streaming Multiprocessor. It is a collection of 32 (NVIDIA) or 64 (AMD) threads that are issued a single instruction at a time and execute it synchronously in SIMD fashion.

34 Correct Answer: B. Tightly coupled Shared Memory Systems (e.g., SMP/UMA systems).

The Snooping Protocol relies on processors constantly "snooping" or monitoring the shared bus that connects all processors and memory. Since all cache transactions are broadcast over this shared medium, any processor can see (snoop on) the memory requests of others, making it highly effective for relatively small, bus-based, shared-memory (UMA/SMP) systems.

35 Correct Answer: A. Two independent variables are accessed by different cores but reside in the same cache line.

False Sharing occurs when unrelated, independent variables that are frequently updated by different cores happen to map to the same cache line. Because the cache coherence protocol operates at the cache line granularity, every write to one variable by one core causes the cache line to be invalidated in the other core's cache, forcing unnecessary and continuous line transfers.

36 Correct Answer: B. It provides uniform, non-blocking bandwidth between all compute nodes.

The Fat Tree topology is highly popular in large HPC clusters (e.g., using InfiniBand or Omni-Path). It is designed to maintain consistent or non-blocking bandwidth capacity throughout the network, particularly at higher levels, ensuring high-speed, uniform communication between any pair of compute nodes.

37 Correct Answer: C. Atomicity

Atomicity (implemented through atomic instructions or locks like mutexes) guarantees that a set of memory operations or instructions is executed completely, without interruption or interference from any other thread or processor. This is essential for maintaining data consistency in shared-memory environments.

38 Correct Answer: B. CPU: Optimizing for Single-Thread Performance; GPU: Optimizing for Parallel Throughput.

CPUs are designed with large caches and complex control units to minimize latency for sequential tasks (high performance for a single thread). GPUs are designed with a large number of simpler cores (ALUs) and execution units to maximize the total amount of parallel work done per second (throughput), often tolerating higher latency for individual operations.

39 Correct Answer: C. Transfer data between I/O devices and main memory without continuous CPU intervention.

DMA is a system capability that allows I/O devices (like network cards or disk controllers) to exchange data directly with main memory. This bypasses the CPU for bulk data transfers, freeing the CPU to perform other tasks and significantly improving system efficiency.

40 Correct Answer: C. Pipelined cryptographic decryption systems where multiple functional units apply different steps to the same data stream.

MISD involves multiple independent instruction streams operating on the same single data stream. While not used for general computing, specialized applications like pipelined signal processing or fault-tolerant control systems can be modeled as MISD, where different processing units (instructions) simultaneously work on the same input data as it passes through the stages (pipeline).

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